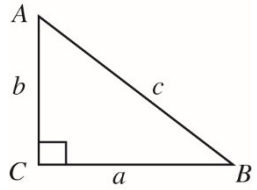
Solve a Right Triangle

Pythagorean theorem: $a^2 + b^2 = c^2$ $\qquad A + B + C = 180°$

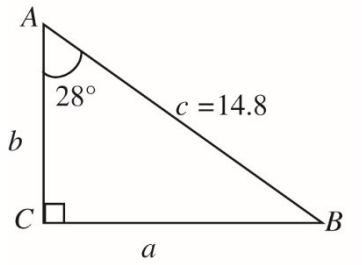


$$\sin\theta = \frac{\text{opp}}{\text{hyp}},\ \cos\theta = \frac{\text{adj}}{\text{hyp}},\ \tan\theta = \frac{\text{opp}}{\text{adj}}$$

$$\csc\theta = \frac{\text{hyp}}{\text{opp}},\ \sec\theta = \frac{\text{hyp}}{\text{adj}},\ \cot\theta = \frac{\text{adj}}{\text{opp}}$$

---

1. Solve the right triangle. Round the lengths of sides to 1 decimal place.

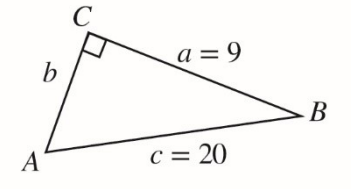


$$B = 180° - 90° - 28° = 62°$$

$$\sin 28° = \frac{a}{14.8} \rightarrow a = 14.8 \sin 28° \approx 6.9$$

$$\sin 62° = \frac{b}{14.8} \rightarrow b = 14.8 \sin 62° \approx 13.1$$

2. Solve the right triangle. Round the measurements to 1 decimal place.



$$b^2 + 9^2 = 20^2$$

$$b^2 = 400 - 81$$

$$b^2 = 319$$

$$b = \sqrt{319} \approx 17.9$$

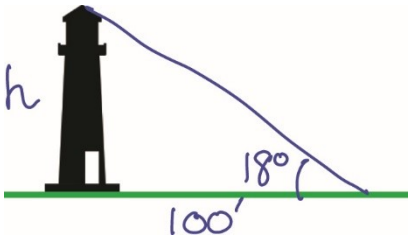
$$\cos B = \frac{9}{20} \rightarrow B = \cos^{-1}\frac{9}{20} \approx 63.3°$$

$$A \approx 180° - 90° - 63.3° = 26.7°$$

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Solve Applications of Right Triangles

3. A surveyor stands on level ground 100 feet from the base of a lighthouse. The measure of the angle from the ground to the top of the lighthouse is 18°. Rounded to the nearest foot, what is the height of the lighthouse?



$$\tan 18° = \frac{h}{100}$$

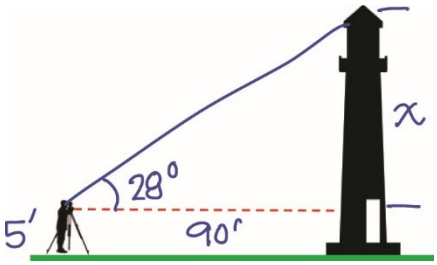
$$h = 100 \tan 18° \approx 32 \text{ feet}$$

4. A surveyor stands on level ground 90 feet from the base of a lighthouse. The measure of the angle from eye level to the top of the lighthouse is 28°. If the surveyor's eye is 5 feet above the ground, what is the height of the lighthouse? Round to the nearest foot.

$$\text{lighthouse height} = x + 5$$

$$\tan 28° = \frac{x}{90} \rightarrow x = 90 \tan 28° \approx 48$$

$$\text{height} \approx 48 + 5 = 53 \text{ feet}$$



5. To determine the height of a window in the top of a lighthouse, two measurements are taken from the ground 90 feet from the base of a lighthouse. The angle of elevation from the ground to the bottom of the window is 36° and the angle of elevation from the ground to the top of the window is 42°.

Find the height of the window to the nearest foot.

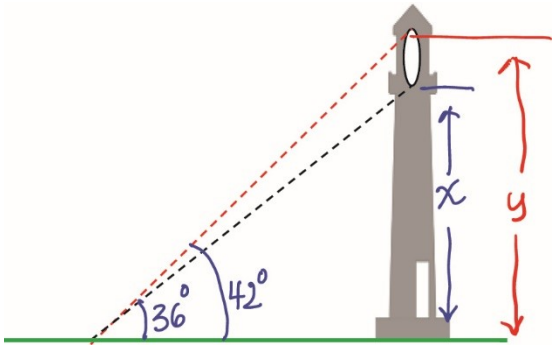
$$\text{smaller triangle} \quad \tan 36° = \frac{x}{90}$$

$$x = 90 \tan 36° \approx 65.3888$$

$$\text{larger triangle} \quad \tan 42° = \frac{y}{90}$$

$$y = 90 \tan 42° \approx 81.0364$$

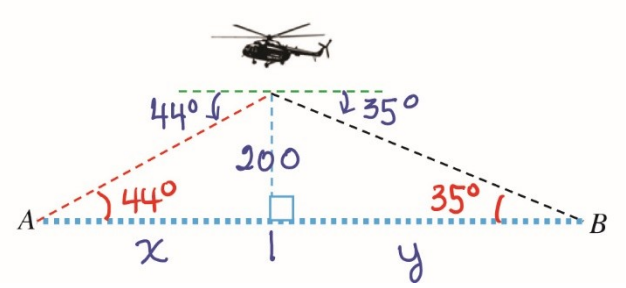
$$\text{window} \approx 81 - 65 = 16 \text{ feet}$$



From the given information, can we determine the height of the lighthouse? No

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6. An observer in a helicopter hovering 200 feet above the sea makes measurements to two buoys at points A and B below. The angles of depression are 44° and 35° respectively. Assuming the observer is in the same vertical plane as the buoys, what is the distance between the two buoys? Round to the nearest foot.



$$\tan 44° = \frac{200}{x}$$

$$x = \frac{200}{\tan 44°} \approx 207$$

$$\tan 35° = \frac{200}{y}$$

$$y = \frac{200}{\tan 35°} \approx 286$$

$$\text{distance} = x + y$$
$$\approx 207 + 286$$
$$= 493 \text{ feet}$$

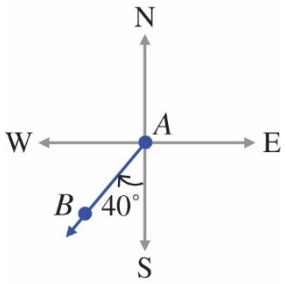
Compute the Bearing of an Object

The term *bearing* is used to convey the angular relationship from one point to another.

The **bearing** from point A to point B is the degree measure of the acute angle formed by ray $\overrightarrow{AB}$ and the north-south line.

A bearing is denoted by the letter N or S (north or south), followed by the measure of the angle in degrees, followed by the letter E or W (east or west).

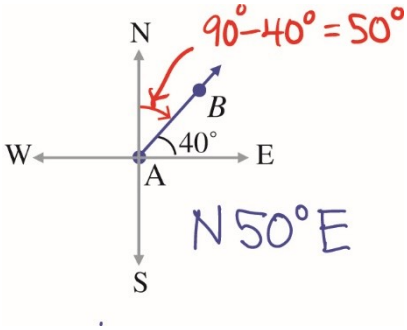
For example, a ship bearing S40°W is heading in a southwesterly direction. Specifically, the ship is traveling 40° to the west of due south.



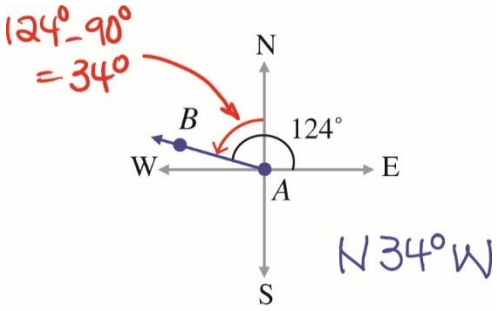
Find the bearing from A to B.

7.

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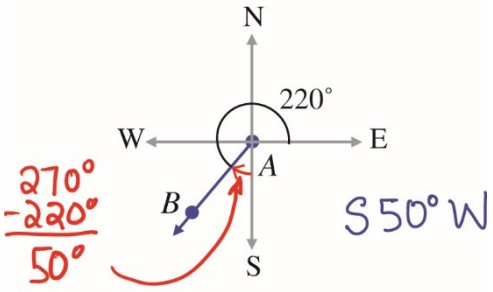


8.

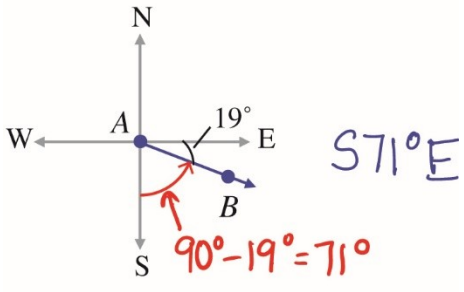


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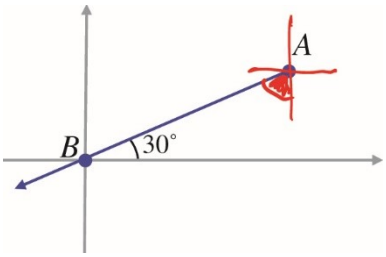
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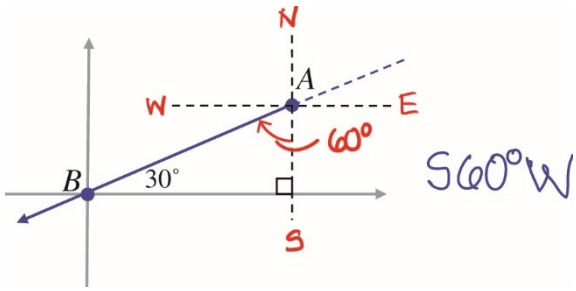
10.



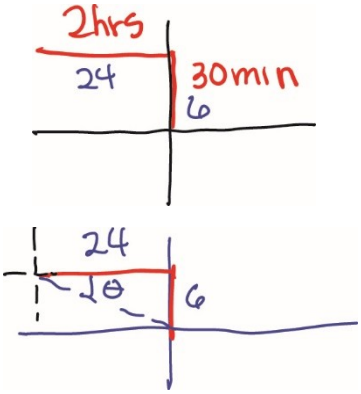
11. Find the bearing from A to B.



Hint: Sketch a coordinate system with origin at point A.



12. A boat leaves a marina and sails north for 30 minutes, then west for 2 hours. If the boat's average speed is 12 miles per hour, find the bearing for the return trip. Round to the nearest degree.

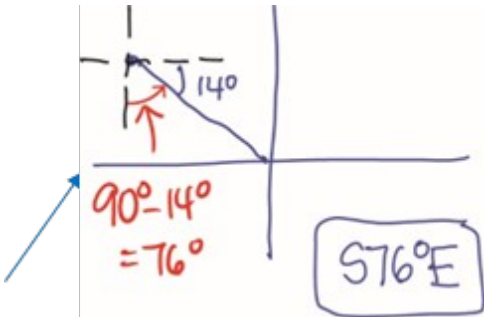


$$2\text{ hr}\cdot\frac{12\text{ mi}}{\text{hr}}=24\text{ mi}$$

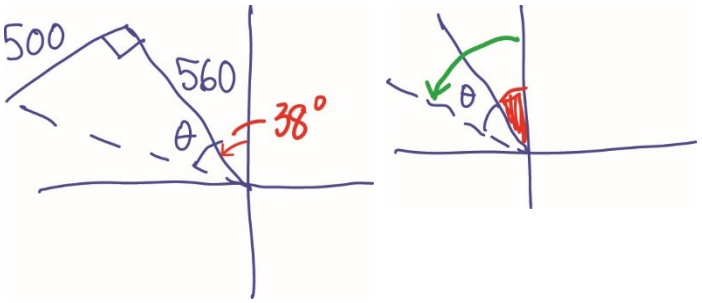
$$\frac{1}{2}\text{ hr}\cdot\frac{12\text{ mi}}{\text{hr}}=6\text{ mi}$$

$$\tan\theta=\frac{6}{24}$$

$$\theta=\tan^{-1}\frac{6}{24}\approx 14°$$



13. A plane leaves an airport at a bearing of N38°W and travels at 280 mph for 2 hours. The plane then makes a 90° left turn and travels 500 miles to its destination. Find the bearing required to locate the plane from the airport. Round to the nearest degree.



$$\frac{280\text{ mi}}{\text{hr}}\cdot 2\text{ hr}=560\text{ mi}$$

$$\tan\theta=\frac{500}{560}$$

$$\theta=\tan^{-1}\frac{500}{560}\approx 42°$$

$38°+42°=80°$ $\boxed{\text{N80°W}}$

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